

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

Callista Lodzinski, Mathilde Muller, and Elizabeth Schuck

Due: 29 May 2026

Test

1 Replication Target

In this project, we aim to replicate the findings of *From the French franc to the euro, is there an economic impact for the CFA franc zone countries? A 'bias-corrected' synthetic control method* by Alassane Diallo and Adama Ba, published in Cogent Economics & Finance in 2024. This paper assesses the impact of pegging the CFA franc to the euro on real GDP per capita in the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CAEMC) countries. The original study uses a set of 35 annual country-level panel data from 1980 to 2019 and applies synthetic control methods to estimate the counterfactual economic outcomes for member countries.

2 Data

Table 1: Variables

Variable	Type	Category	(First) Page Ref	Possible S
Real GDP per capita	Continuous	Outcome	6	WDI, Penn
Lagged GDP per capita	Continuous	Lagged outcome	10	WDI, Penn
Gross Fixed Capital Formation (% GDP)	Continuous	Capital	10	
Government expenditure (% GDP)	Continuous	Fiscal	10	WDI, Penn
Official Development Assistance (% GDP)	Continuous	External	10	Source not
Foreign Development Investment (% GDP)	Continuous	External	10	WDI
Employment rate	Continuous	Labour	10	WDI
Agriculture (% GDP)	Continuous	Structure	10	WDI, Penn
Industry (% GDP)	Continuous	Structure	10	WDI, Penn
Governance / institutional quality	Index	Institutions	10	Polity5 Dat
Political regime characteristics	Categorical/Index	Institutions	10	Polity5 Dat
Inflation	Continuous	Macro controls	5	WDI, Penn
Euro peg dummy	Binary	Treatment		Authors
Synthetic GDP per capita	$\tilde{Y}_{it}$	Continuous	Constructed	
Treatment effect	τ_{it}	Continuous	Constructed	

3 Appendix 1

Table 2: Treated countries and regional classification

Country	ISO3	CFA monetary union	Included in SCM sample	Notes
Benin	BEN	WAEMU	Yes	Original CFA member
Burkina Faso	BFA	WAEMU	Yes	Original CFA member
Côte d'Ivoire	CIV	WAEMU	Yes	Original CFA member
Mali	MLI	WAEMU	Yes	Rejoined WAEMU in 1984
Niger	NER	WAEMU	Yes	Original CFA member
Senegal	SEN	WAEMU	Yes	Original CFA member
Togo	TGO	WAEMU	Yes	Original CFA member
Cameroon	CMR	CEMAC	Yes	Original CFA member
Central African Republic	CAF	CEMAC	Yes	Original CFA member
Chad	TCD	CEMAC	Yes	Original CFA member
Republic of Congo	COG	CEMAC	Yes	Original CFA member
Gabon	GAB	CEMAC	Yes	Original CFA member
Equatorial Guinea	GNQ	CEMAC	Yes	Joined CFA franc zone in 1985
Guinea-Bissau	GNB	WAEMU	No	Joined WAEMU in 1997; excluded

Table 3: Variable construction in the original paper, updated replication, and trade extension

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Real GDP per capita / selected GDP per capita	gdp_selected; wdi_gdp_pc_current	Main GDP outcome in replication	World Development Indicators	The original paper studies real GDP per capita. In the replication, alternative GDP measures were compared and WDI GDP per capita, current US dollars, was selected as the closest empirical match to the paper's reported results.	Main divergence from the original paper; The exact GDP series in the paper could not be reproduced unambiguously, considering numerous WDI and PWT indicators.
Agriculture value added	agriculture	SCM predictor	WDI and FAOSTAT	Agriculture, forestry, and fishing, value added as a percentage of GDP. WDI indicator: NV.AGR.TOTL.ZS . Alternative agriculture series were used to fill major gaps.	Retained in the final predictor set.
Industry value added	industry	Original/intended SCM predictor	WDI	Industry including construction, value added as a percentage of GDP. Main WDI indicator: NV.IND.TOTL.ZS .	Included in the original paper's predictor set but removed from the final replication because of missingness and data-quality problems.
Government expenditure share	govt_share	SCM predictor	Penn World Table	Government consumption share from PWT, using csh.g .	Corresponds to the government expenditure control described in the original paper.
Investment share / gross fixed capital formation	invest_share	SCM predictor	Penn World Table	Investment or capital-formation share from PWT, using csh.i .	Corresponds to the paper's gross fixed capital formation control.
Official development assistance	oda_share	SCM predictor	World Development Indicators	Main WDI series used include net ODA received. DT.ODA.ODAT.GD.ZS .	ODA as a share of national income.

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Foreign direct investment	fdi	SCM predictor	World Development Indicators	Foreign direct investment, net inflows as a percentage of GDP. WDI indicator: BX.KLT.DINV.WD.GD.ZS .	Retained in the final predictor set.
Labour / employment	labour	SCM predictor	Penn World Table	Employment as a share of population, constructed as $100 * \text{emp} / \text{pop}$.	Corresponds to the original paper's labour factor, described as the proportion of the population in work.
Political regime / institutions	polity2	SCM predictor	Polity5	Polity2 institutional score, ranging from autocracy to democracy (-10 to +10).	Interpolation and extrapolation were applied where needed to obtain usable pre-treatment panels.
1994 CFA devaluation dummy	devaluation_1994	Original control / historical adjustment	Constructed	Dummy variable intended to capture the January 1994 CFA franc devaluation.	The original paper mentions this control. In the replication code, it was not retained as a standard SCM predictor because it has no useful cross-donor variation.
Trade openness	trade_openness	Trade-extension outcome	World Development Indicators	Exports plus imports of goods and services as a percentage of GDP. WDI indicator: NE.TRD.GNFS.ZS .	Used to test whether the euro peg changed aggregate trade exposure.
Exports of goods and services	exports_gdp	Trade-extension outcome	World Development Indicators	Exports of goods and services as a percentage of GDP. WDI indicator: NE.EXP.GNFS.ZS .	Used to examine export intensity.
Imports of goods and services	imports_gdp	Trade-extension outcome	World Development Indicators	Imports of goods and services as a percentage of GDP. WDI indicator: NE.IMP.GNFS.ZS .	Used to examine import intensity.
Exports to EU share	XEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times exports to the EU divided by exports to the world.	Measures export composition toward the EU.

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Imports from EU share	MEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times imports from the EU divided by imports from the world.	Measures import composition toward the EU.

Table 4: Main GDP replication design

Element	Implementation in the reconstructed replication
Research design	Country-level synthetic control method, estimated separately for each treated CFA country.
Treatment	Shift from French franc anchor to euro anchor.
Treatment year	2002.
Pre-treatment period	1980–2001.
Post-treatment period	2002–2019.
Treated units	13 CFA countries: seven WAEMU countries and six CEMAC countries.
Main outcome	WDI GDP per capita in current US dollars.
Donor pool	Updated donor pool of non-CFA countries, based on the original donor list and augmented with additional fixed-exchange-rate candidates.
Predictors	Agriculture, government share, investment share, ODA, FDI, labour, and polity score.
Lagged outcome predictors	GDP values in selected pre-treatment years: 1980, 1990, 1995, and 2001.
Excluded predictor	Industry, dropped because of missingness and data-quality concerns.
Inference	In-space placebo tests using donor countries as placebo-treated units.
Interpretation	Transparent reconstructed replication, not exact reproduction of the authors’ original dataset.

Table 5: Variable construction in the reconstructed GDP replication

Variable	Code name	Source	Construction and notes
GDP per capita	<code>gdp_selected;</code> <code>wdi_gdp_pc_current</code>	WDI	Main outcome. WDI GDP per capita in current US dollars was selected because the GDP values in the original paper could not be matched to any standard WDI or PWT GDP series.
Lagged GDP	<code>gdp_selected</code> in selected years	Constructed from selected GDP outcome	Used as special SCM predictors for 1980, 1990, 1995, and 2001.
Agriculture value added	<code>agriculture</code>	WDI; FAOSTAT fallback	WDI agriculture value added share, supplemented with FAOSTAT where WDI observations were missing. Retained in final SCM predictor set.
Industry value added	<code>industry</code>	WDI	Initially considered because it appears in the original specification, but excluded from final SCM because of missingness and data-quality problems.
Government share	<code>govt_share</code>	Penn World Table	Constructed from PWT government consumption share. Retained as SCM predictor.
Investment share	<code>invest_share</code>	Penn World Table	Constructed from PWT investment share. Retained as SCM predictor.
Official development assistance	<code>oda_share</code>	WDI	Constructed from WDI ODA data. The denominator used in the project code should be interpreted cautiously and is documented as a reconstruction choice.
Foreign direct investment	<code>fdi</code>	WDI	WDI foreign direct investment, net inflows as a share of GDP. Retained as SCM predictor.
Labour	<code>labour</code>	Penn World Table	Constructed as employment divided by population, multiplied by 100. Retained as SCM predictor.
Political institutions	<code>polity2</code>	Polity5	Polity score. Missing values were filled using documented imputation rules where needed.

Variable	Code name	Source	Construction and notes
Inflation	inflation and related deflator series	WDI and PWT	Used only for descriptive appendix tables. Published inflation-table values were closer to PWT deflator-based values than to WDI inflation indicators.

Table 6: Main replication challenges and modelling decisions

Issue	Problem	Replication decision
GDP variable	The paper describes the outcome as real GDP per capita, but the reported values could not be matched to standard WDI or PWT GDP series.	Tested multiple GDP alternatives and selected WDI GDP per capita in current US dollars as the closest publicly reproducible match.
Missing-data treatment	The paper does not document whether missing values were dropped, interpolated, extrapolated, or filled from other sources.	Saved both unimputed and imputed panels and applied documented imputation rules before SCM estimation.
Agriculture predictor	WDI agriculture data contained gaps for some country-years.	Supplemented missing WDI agriculture observations with FAO-STAT data where available.
Industry predictor	Industry had substantial missingness and data-quality issues.	Dropped industry from the final SCM predictor set.
Donor pool	The original donor-pool reporting was not fully reproducible and some donor countries created data limitations.	Used an explicitly defined updated donor pool, retaining most original donors and adding Turkmenistan, Guyana, and Iraq.
ODA construction	The original paper does not clearly document the ODA construction.	Constructed ODA from WDI data and treated the variable as a documented reconstruction choice.
Inflation tables	The published inflation tables did not match the stated WDI source closely.	Replicated the descriptive inflation tables separately and placed them in the appendix.
Inference	The original paper uses bias-corrected SCM; the replication uses standard Synth SCM in R.	Interpreted results as a reconstructed SCM replication and used donor-placebo tests for inference.

Table 7: Missing-data treatment in the reconstructed panel

Variable or issue	Method used	Notes
Polity score	Nearest-observation fill	Applied generally to fill missing institutional observations.
Agriculture	FAOSTAT supplement; nearest-observation fill	FAOSTAT was used to fill missing WDI agriculture values where possible. Remaining gaps were handled through the imputation plan.
Industry	Nearest-observation fill in data construction; excluded from final SCM	Although industry was included in the broader data construction process, it was dropped from final SCM because data quality remained problematic.
ODA	Country-specific and general nearest-observation fill	Several donor countries required endpoint filling. ODA construction remains a reconstruction choice because the original paper does not clearly document the denominator.
FDI	Zero-fill for selected early gaps; nearest-observation fill generally	Zero-filling was used for selected cases where missing early values were treated as approximately zero; other gaps were filled using nearest observations.
Labour	Country-specific and general nearest-observation fill	Used for countries with early, late, or broad labour-data gaps.

Variable or issue	Method used	Notes
Government share	Nearest-observation fill	Constructed from PWT and filled where necessary.
Investment share	Nearest-observation fill	Constructed from PWT and filled where necessary.
GDP variables	Nearest-observation fill; selected linear interpolation	Several GDP alternatives were constructed and filled during data preparation, though the final outcome was WDI GDP per capita in current US dollars.
Donor eligibility	Donor screening before SCM	Donors entirely missing at least one active predictor over the pre-treatment period were excluded from the usable SCM donor pool.

Table 8: Original paper specification and reconstructed replication

Element	Original paper	Reconstructed replication
Outcome	Described as real GDP per capita.	WDI GDP per capita in current US dollars, selected after testing alternative GDP series.
SCM implementation	Bias-corrected SCM using Stata implementation.	Standard SCM using R <code>Synth</code> .
Treated countries	CFA-zone countries; reporting is not fully consistent across tables.	13 treated CFA countries; Guinea-Bissau excluded from main SCM sample.
Treatment year	2002.	2002.
Pre-treatment period	1980–2001 or 1980–2002 depending on wording.	1980–2001.
Donor pool	Published donor pool, with some ambiguity in reporting.	Explicit updated donor pool defined in shared constants.
Predictors	Sectoral, macroeconomic, institutional, and lagged-GDP predictors, including industry.	Agriculture, government share, investment share, ODA, FDI, labour, polity score, and lagged GDP; industry excluded.
Missing data	Not documented.	Documented imputation plan; both unimputed and imputed panels saved.
Inflation table	Described as WDI-based.	Replicated separately; values appear closer to PWT deflator-based series.

4 Synthetic control method and analysis pipeline

5 Replication strategy and data

The replication follows the original paper’s country-level synthetic control design. For each treated CFA franc country, we estimate a counterfactual GDP path using a weighted average of non-CFA donor countries, and then compare the observed post-2002 GDP path with this synthetic counterfactual. The treatment year is 2002, corresponding to the shift from the French franc anchor to the euro anchor. The pre-treatment period is 1980–2001 and the post-treatment period is 2002–2019. The main treated sample contains 13 CFA countries across WAEMU and CEMAC. Further details on the sample, donor pool, variable construction, and missing-data treatment are reported in Appendix ??, Table 3, and Table 7.

Let Y_{it} denote GDP per capita for country i in year t . For a treated country $i = 1$, the unobserved counterfactual outcome after treatment is approximated using a weighted average of J donor countries:

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad (1)$$

where the donor weights satisfy the standard synthetic control constraints:

$$w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1. \quad (2)$$

The weights are chosen to make the synthetic control resemble the treated country as closely as possible before treatment. Let X_1 denote the vector of pre-treatment characteristics for the treated country, and let X_0 denote the corresponding matrix for the donor countries. The donor-weight vector W is selected to minimise the distance between the treated country and the weighted donor average:

$$\min_W (X_1 - X_0 W)' V (X_1 - X_0 W), \quad (3)$$

subject to the constraints in Equation 2. The matrix V determines the relative importance assigned to each predictor in matching the treated country to the donor pool. In the reconstructed specification, the predictors include agriculture, government share, investment share, official development assistance, foreign direct investment, labour, institutional quality, and selected pre-treatment GDP values. Industry is excluded from the final specification because of missingness and data-quality concerns, as discussed in Appendix ??.

The estimated treatment effect is the post-treatment gap between the observed outcome and the synthetic counterfactual:

$$\hat{\alpha}_{1t} = Y_{1t} - \hat{Y}_{1t}^N = Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}, \quad t \geq 2002. \quad (4)$$

A positive value of $\hat{\alpha}_{1t}$ indicates that actual GDP per capita was above the synthetic counterfactual, while a negative value indicates that actual GDP per capita was below the counterfactual. The interpretation of these gaps depends on the quality of the pre-treatment fit: if the synthetic control does not reproduce the treated country's pre-2002 path closely, the post-treatment gap is less informative.

The replication is best understood as a reconstructed replication rather than a mechanical reproduction of the original paper. Two issues were especially important. First, the GDP series used in the original paper could not be identified exactly from the published documentation. Second, the paper does not report how missing observations were handled, even though missing outcomes or predictors can affect donor eligibility, predictor averages, and estimated synthetic controls. These issues are discussed in Appendix ??; the final variable definitions are summarised in Table 3, and the missing-data procedures are summarised in Table 7.

The analysis proceeds in four steps:

Step	Description
1	The country-year panel is reconstructed from public datasets.
2	Variables are harmonised and missing values are handled using documented rules.
3	One synthetic control is estimated for each treated country, producing donor weights, predictor-balance statistics, actual-versus-synthetic GDP paths, and post-treatment gaps.
4	In-space placebo tests are used to compare the treated-country gaps with gaps obtained when the same procedure is applied to donor countries as placebo-treated units.

Several additional reporting issues in the original paper also shaped the replication strategy. These include donor-pool ambiguity, inconsistent country naming, unclear variable definitions, inflation-source inconsistencies, table-label errors, and numerical values that are difficult to reconcile with the stated data structure. These issues do not invalidate the paper's research question, but they reduce confidence that the published tables provide sufficient information for an exact replication. The purpose of the replication is therefore to evaluate the robustness of the paper's broad conclusions using a transparent, reproducible data construction and estimation procedure.

Table 9: Estimated synthetic control weights by treated country

Donor country	Benin	Burkina Faso	Côte d'Ivoire	Mali	Niger	Senegal	Togo	Cameroon	CAR	CI
Bangladesh	.771	.901	.353	.946	.757	.356	0	.072	.785	.9
Barbados	.002	0	0	0	0	0	0	0	0	0
Bhutan	.124	0	0	0	0	0	0	.053	0	0
Bolivia	0	0	.093	0	0	.261	.011	.002	0	0
Botswana	0	0	0	0	0	0	0	0	0	0
Cabo Verde	0	0	0	0	0	0	0	0	0	0
Dominica	0	0	0	0	0	0	0	0	0	0
Ecuador	.001	0	.110	0	0	0	0	.013	0	0
Eswatini	0	0	0	.008	0	0	0	0	0	0
Grenada	0	0	0	0	0	0	0	0	0	0
Lao PDR	.094	.097	.120	.040	.236	.047	.301	.017	.191	.0
Lesotho	0	0	0	0	0	.203	.486	0	0	0
Mauritius	0	0	0	0	0	0	0	.083	0	0
Morocco	0	0	.225	0	0	0	0	0	0	0
Namibia	0	0	.074	0	0	.090	0	0	0	0
St. Kitts and Nevis	0	0	0	0	0	0	0	0	0	0
St. Lucia	0	0	0	0	0	0	0	0	0	0
Seychelles	0	0	0	0	0	0	0	0	0	0
Turkmenistan	0	0	0	0	0	0	.187	.738	0	0
Guyana	0	0	0	0	0	0	0	0	0	0
Iraq	.008	.002	.026	.006	.008	.043	.015	.022	.023	0

6 Replication Tables

Table 10: The average of the predictors of per-capita GDP for franc-CFA zone countries

country	avg_agriculture	avg_government	avg_oda	avg_fdi	avg_labour	avg_
Benin	29.961610	0.1633115	54778167	0.4055924	35.62389	
Synthetic Benin	24.046460	0.1368013	279752921	0.7386912	35.40573	
Burkina Faso	27.009303	0.1292807	137699348	0.6938516	40.17606	
Synthetic Burkina Faso	24.113678	0.1115558	324146189	0.7302257	34.87846	
Côte d'Ivoire	21.244944	0.1547087	58697793	1.1245368	31.40414	
Synthetic Côte d'Ivoire	18.671979	0.1611628	152098989	1.5637902	35.37862	
Mali	31.924485	0.2248145	130880367	1.4313488	30.88820	
Synthetic Mali	23.398921	0.1071283	338448055	0.5959800	33.96791	
Niger	36.071288	0.2076225	133696937	2.2883942	41.15699	
Synthetic Niger	25.420032	0.1262732	279822237	1.1014510	36.74512	
Senegal	16.892927	0.1675746	72103592	1.3178785	33.58079	
Synthetic Senegal	16.282669	0.1718504	152571094	1.8662555	35.34647	
Togo	21.379832	0.1584141	31225374	1.4919279	40.15292	
Synthetic Togo	18.256600	0.2420278	27556161	3.0518995	39.29024	
Cameroon	19.749879	0.1512596	53866423	1.2173302	39.96706	
Synthetic Cameroon	18.949749	0.2934358	32124099	3.7124711	43.11401	
Central African Republic	37.435338	0.1980131	55865711	0.8982344	39.26813	
Synthetic Central African Republic	24.722553	0.1231565	290418608	0.9711322	35.87492	
Chad	34.548047	0.0753845	72635846	4.1151560	38.32956	
Synthetic Chad	23.836711	0.1082252	334233918	0.6479046	34.47783	
Republic of Congo	7.863290	0.3325782	14367386	5.8508828	33.37802	
Synthetic Republic of Congo	17.744385	0.2940493	11537614	3.8061232	43.27624	
Gabon	6.291786	0.1734423	1457547	2.4415752	30.06796	
Synthetic Gabon	6.789081	0.1871184	43748213	2.7948043	30.81468	
Equatorial Guinea	6.788717	0.3130109	7265594	17.4621540	33.56924	
Synthetic Equatorial Guinea	23.841548	0.3014595	6797661	0.6347582	38.52312	

Table 11: CFA franc zone countries annual inflation rate: WAEMU

country	1980–2001	2002–2021
Benin	7.5270	2.1268
Burkina Faso	4.3427	2.2660
Côte d'Ivoire	8.5558	2.1842
Guinea-Bissau	39.2343	2.8921
Mali	8.6642	3.5490
Niger	6.0235	2.8630
Senegal	5.1354	1.8448
Togo	6.2716	2.0069
Average inflation for the WAEMU zone	10.7193	2.4666
Average inflation without Guinea-Bissau	6.6457	2.4058

Table 12: CFA franc zone countries annual inflation rate: CAEMC

country	1980–2001	2002–2021
Cameroon	7.0840	1.8268
Central African Republic	6.5481	2.9430
Chad	5.2621	5.4660
Republic of Congo	7.3655	6.1262
Equatorial Guinea	11.0392	6.7586
Gabon	7.4425	3.7486
Average inflation for the CAEMC zone	7.4569	4.4782

Table 13: Non-CFA franc zone countries mean annual inflation rate

country	1980–2001	2002–2021
Angola	550.9314	32.1383
Burundi	10.3255	9.2785
Congo, Dem. Rep.	1810.0588	16.3427
Ethiopia	5.2079	13.5134
Gambia, The	14.5303	10.1870
Ghana	36.1786	21.0285
Guinea	16.1342	11.2597
Kenya	11.8579	8.5152
Madagascar	18.4232	8.3332
Malawi	23.6956	18.1634
Nigeria	30.4076	13.5749
Sao Tome and Principe	31.6514	10.6113
Sierra Leone	51.3708	11.1235
Sudan	50.6410	37.8117
Tanzania	21.8489	7.4472
Uganda	53.1089	8.7408
Zambia	51.1202	12.4355
Zimbabwe	-1.6381	71.6617
Average inflation	154.7697	17.8981

Table 14: WAEMU countries' mean GDP per capita

country	Mean GDP pc 1990-2001	Mean GDP pc 2002-2019
Benin	384.662	966.377
Burkina Faso	273.221	601.976
Côte d'Ivoire	926.172	1588.214
Guinea-Bissau	538.662	603.132
Mali	345.299	763.173
Niger	275.470	446.159
Senegal	747.919	1210.000
Togo	455.380	701.218
Average for the WAEMU area	493.348	860.031
Average without Guinea-Bissau	486.875	896.731

Table 15: CAEMC countries' mean GDP per capita

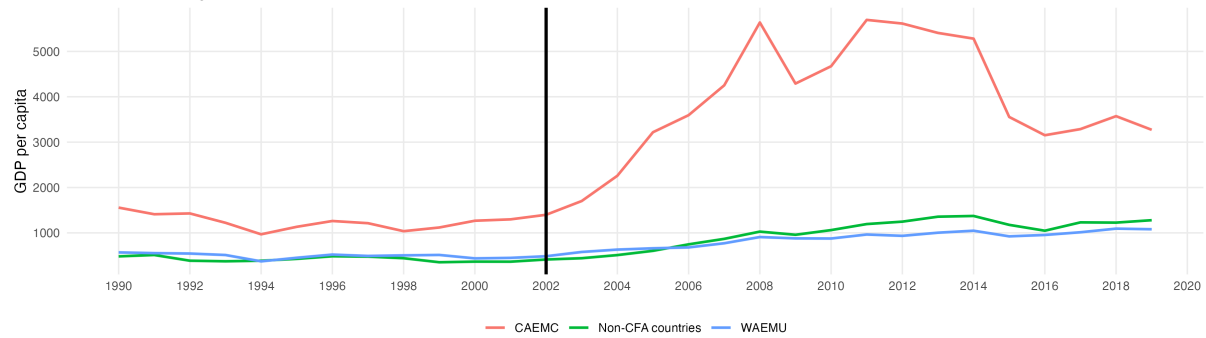
country	Mean GDP pc 1990-2001	Mean GDP pc 2002-2019
Cameroon	881.576	1368.350
Central African Republic	333.161	398.380
Chad	222.043	934.540
Republic of Congo	911.425	2430.081
Equatorial Guinea	622.261	10685.135
Gabon	4487.668	7472.441
Average for the CAEMC zone	1243.022	3881.488

Table 16: Non-CFA countries' mean GDP per capita

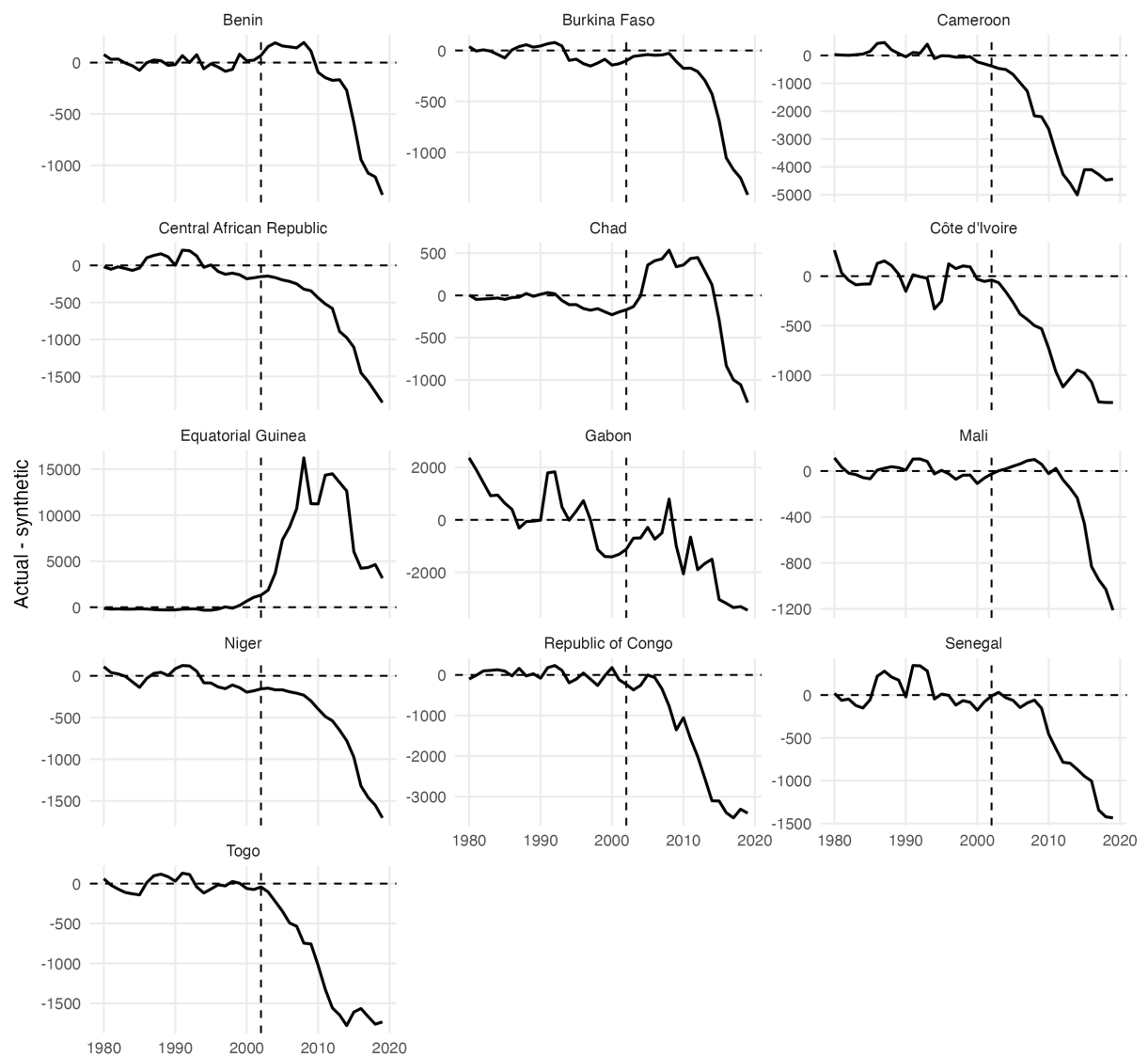
country	Mean GDP pc 1990-2001	Mean GDP pc 2002-2019
Angola	556.462	3364.635
Burundi	160.535	199.858
Congo, Dem. Rep.	187.784	355.273
Ethiopia	163.612	403.230
Gambia, The	571.049	646.718
Ghana	361.406	1350.937
Guinea	640.797	685.543
Kenya	379.630	1120.579
Madagascar	291.122	433.348
Malawi	260.896	519.866
Nigeria	915.649	2001.924
Sao Tome and Principe	753.382	1174.027
Sierra Leone	195.936	730.571
Sudan	541.562	1058.802
Tanzania	311.399	748.458
Uganda	236.552	627.725
Zambia	402.437	1213.198
Zimbabwe	666.466	1136.475
Average	422.037	987.287

7 Replication Figures

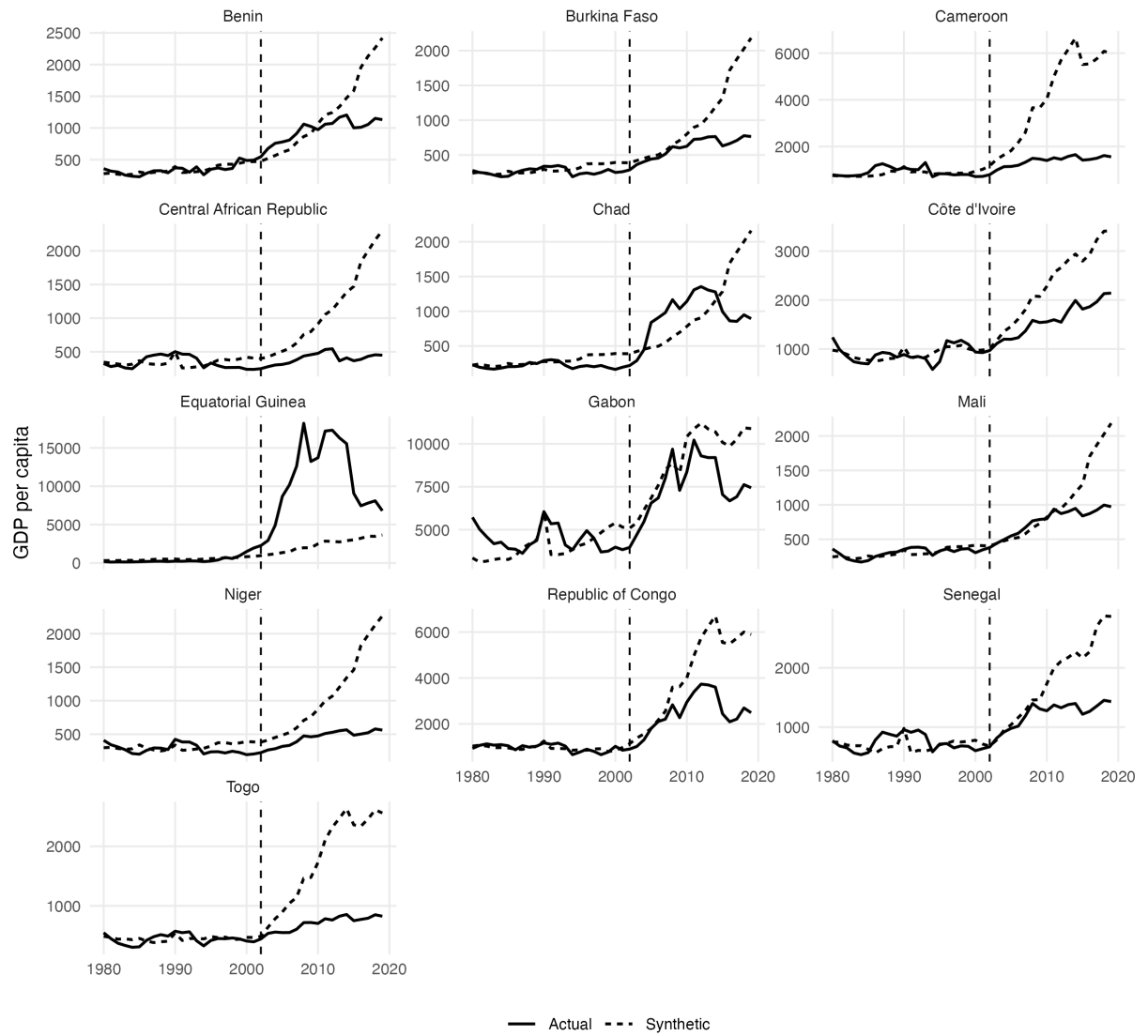
Evolution of GDP per capita
GDP variable: wdi_gdp_pc_current

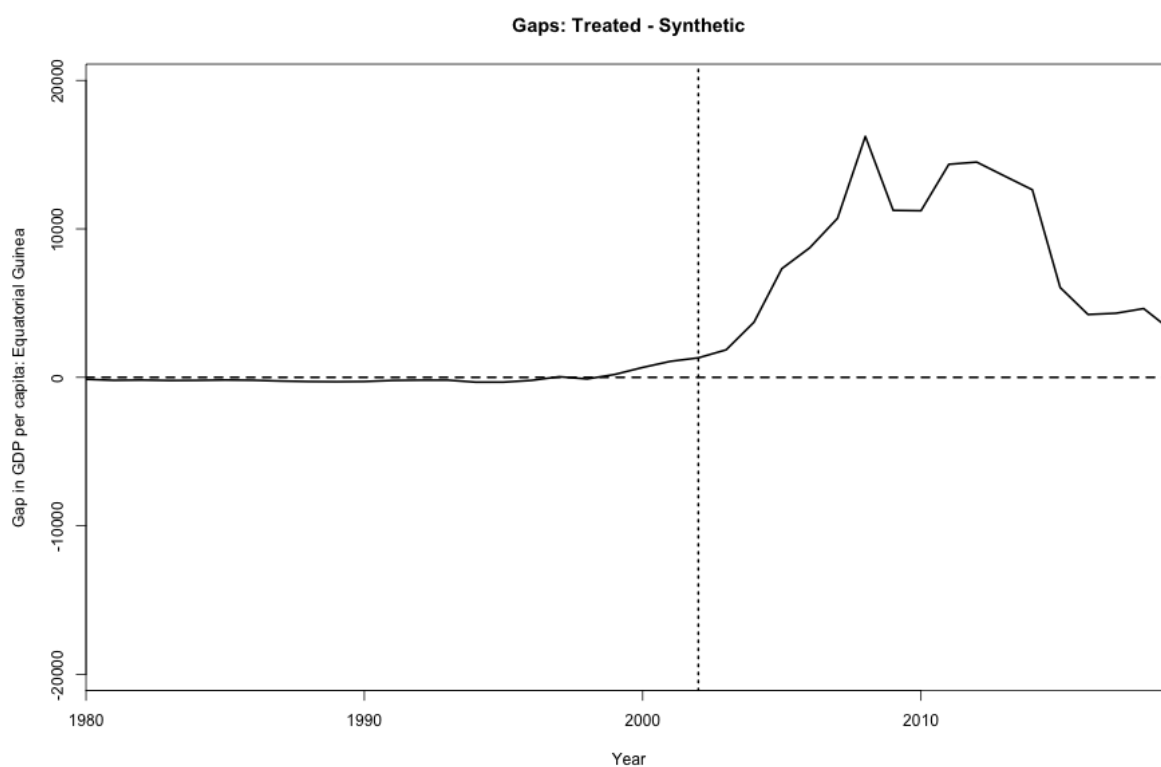
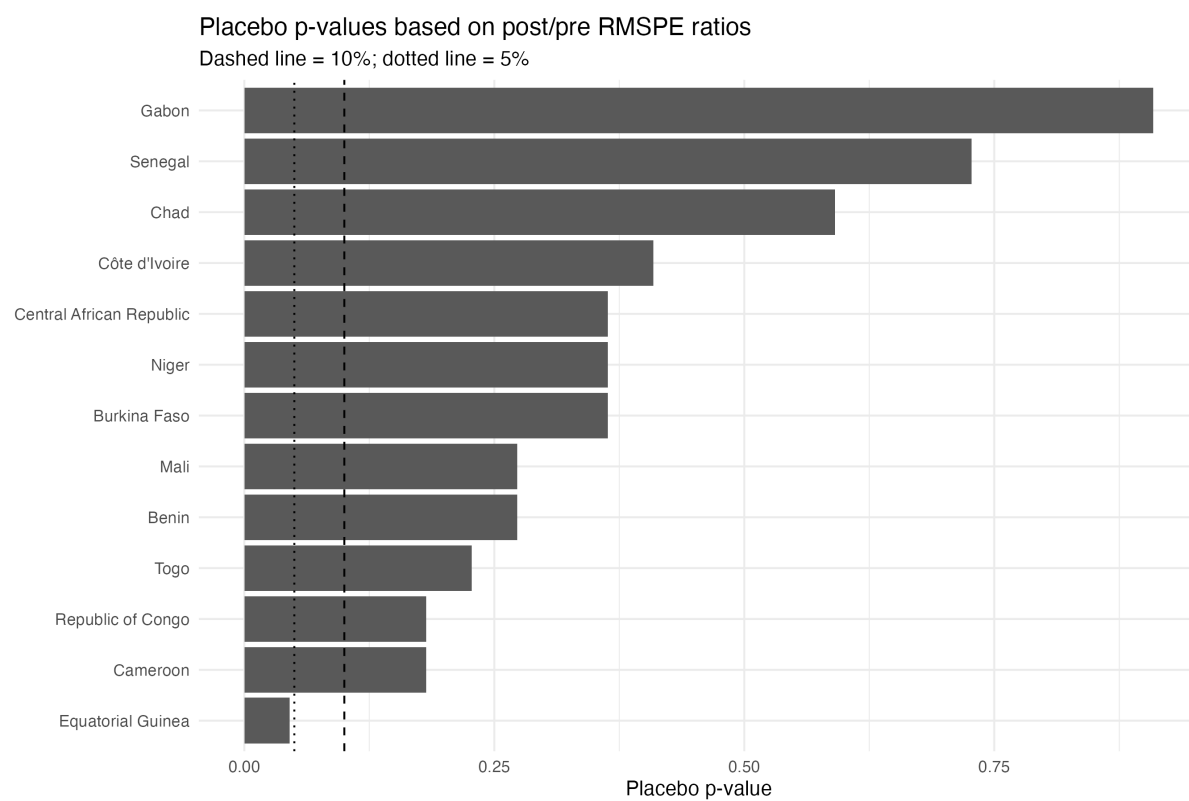


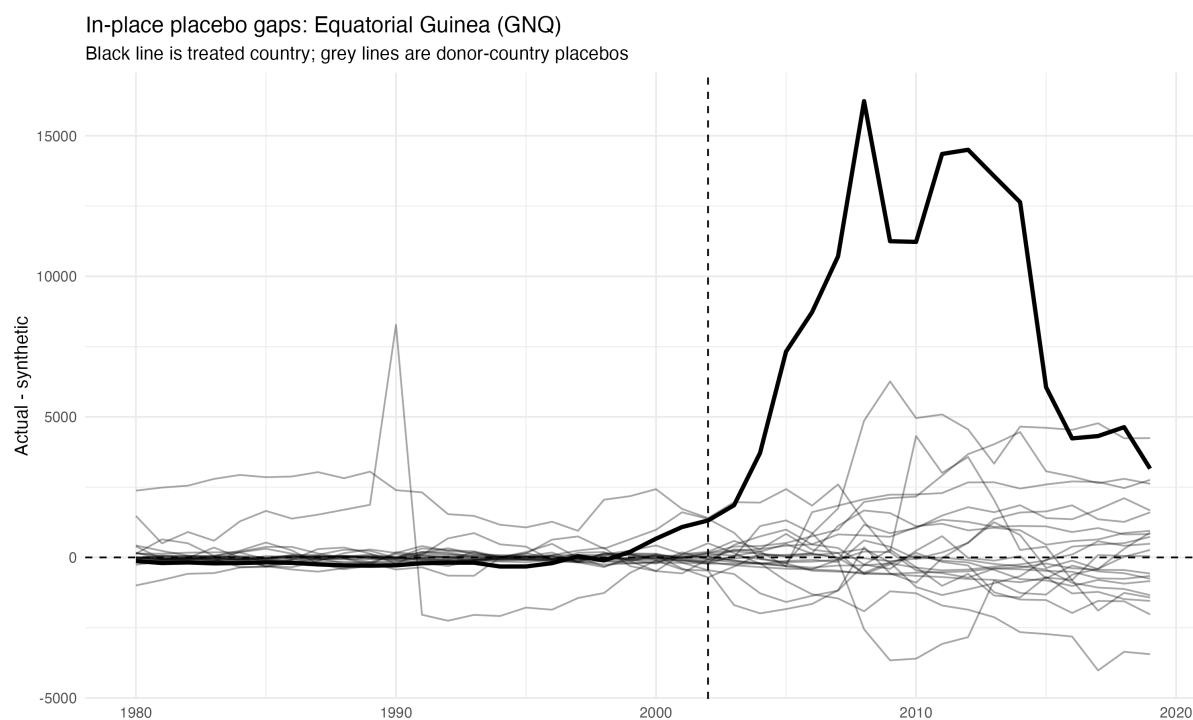
GDP per capita gaps: WAEMU and CAEMC countries
Outcome variable: WDI GDP per capita (current USD)



Actual and synthetic GDP per capita: WAEMU and CAEMC countries
Outcome variable: WDI GDP per capita (current USD)







8 Extension Tables

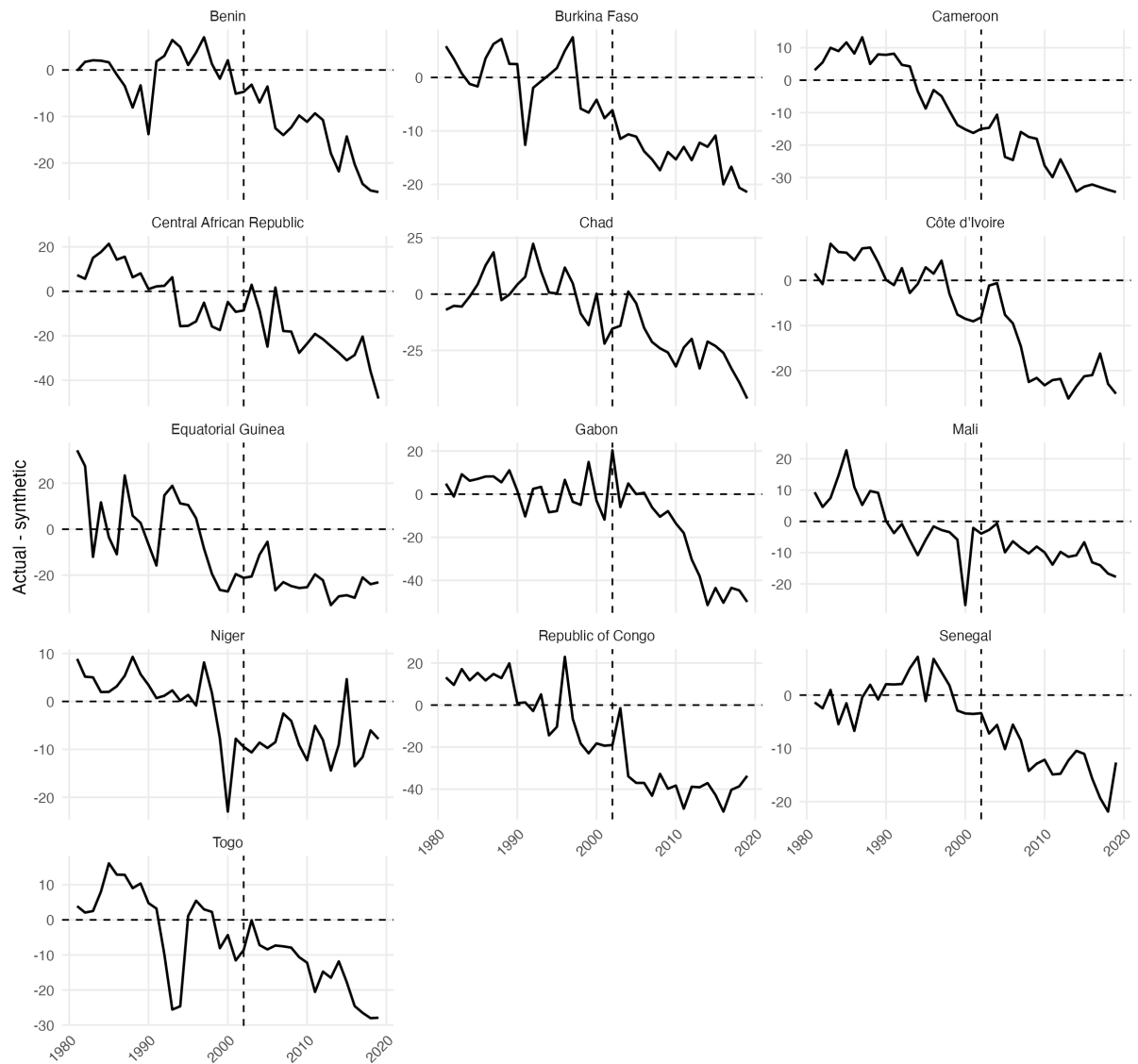
Table 17: Extension SCM summary for trade outcomes

outcome	treated_iso3c	treated_country	pre_rmspe	post_rmspe	rmspe_ratio	avg_post_gap	avg_abs_post_gap	outcome_label
MEU	BEN	Benin	4.740438	15.617041	3.2944296	-13.8408334	13.840833	Imports from EU (% of total imports)
MEU	BFA	Burkina Faso	5.205163	14.847303	2.8524186	-14.3512625	14.351263	Imports from EU (% of total imports)
MEU	CAF	Central African Republic	12.029886	24.439553	2.0315698	-21.1949729	21.714439	Imports from EU (% of total imports)
MEU	CIV	Côte d'Ivoire	5.132861	18.955975	3.6930623	-17.1880613	17.188061	Imports from EU (% of total imports)
MEU	CMR	Cameroon	9.125855	26.179158	2.8686799	-25.0141587	25.014159	Imports from EU (% of total imports)
MEU	COG	Republic of Congo	14.375437	37.836069	2.6319943	-36.2932688	36.293269	Imports from EU (% of total imports)
MEU	GAB	Gabon	7.555036	30.751044	4.0702711	-21.5499129	24.434454	Imports from EU (% of total imports)
MEU	GNQ	Equatorial Guinea	17.363994	23.870283	1.3747001	-23.0139310	23.013931	Imports from EU (% of total imports)
MEU	MLI	Mali	10.233356	10.643753	1.0401039	-9.6770227	9.677023	Imports from EU (% of total imports)
MEU	NER	Niger	7.016746	9.172159	1.3071812	-8.0933944	8.610415	Imports from EU (% of total imports)
MEU	SEN	Senegal	3.668794	12.681645	3.4566250	-11.8079709	11.807971	Imports from EU (% of total imports)
MEU	TCD	Chad	10.329470	25.724796	2.4904275	-23.1336859	23.262100	Imports from EU (% of total imports)
MEU	TGO	Togo	10.970151	16.456914	1.5001538	-14.3598300	14.359830	Imports from EU (% of total imports)
XEU	BEN	Benin	20.203927	16.245284	0.8040657	-16.0536464	16.053646	Exports to EU (% of total exports)
XEU	BFA	Burkina Faso	8.860764	24.476528	2.7623495	-21.8872756	23.559216	Exports to EU (% of total exports)
XEU	CAF	Central African Republic	22.332967	16.361209	0.7326035	-3.7435440	13.077108	Exports to EU (% of total exports)
XEU	CIV	Côte d'Ivoire	6.277816	14.528041	2.3141872	-13.8755376	13.875538	Exports to EU (% of total exports)
XEU	CMR	Cameroon	14.318287	6.591509	0.4603560	-1.0068068	5.260558	Exports to EU (% of total exports)
XEU	COG	Republic of Congo	13.225655	13.642517	1.0315191	-10.4144214	12.755337	Exports to EU (% of total exports)
XEU	GAB	Gabon	15.327433	12.923145	0.8431383	-11.8499627	11.849963	Exports to EU (% of total exports)
XEU	GNQ	Equatorial Guinea	29.864246	32.023694	1.0723088	-30.8151028	30.815103	Exports to EU (% of total exports)
XEU	MLI	Mali	14.845509	26.343900	1.7745367	-25.8672286	25.867229	Exports to EU (% of total exports)
XEU	NER	Niger	12.137954	20.813455	1.7147416	-18.4445869	18.444587	Exports to EU (% of total exports)
XEU	SEN	Senegal	5.393509	24.955852	4.6270159	-24.2435584	24.243558	Exports to EU (% of total exports)
XEU	TCD	Chad	16.434477	49.902716	3.0364652	-47.2601214	47.460121	Exports to EU (% of total exports)
XEU	TGO	Togo	17.258511	16.784060	0.9725092	-16.1284693	16.128469	Exports to EU (% of total exports)
exports_gdp	BEN	Benin	3.361911	6.417701	1.9089446	-6.0602666	6.060267	Exports of goods and services (% of GDP)
exports_gdp	BFA	Burkina Faso	2.598701	6.207541	2.3887094	-5.9584216	5.958422	Exports of goods and services (% of GDP)
exports_gdp	CAF	Central African Republic	2.358274	13.509221	5.7284368	-12.4497160	12.449716	Exports of goods and services (% of GDP)
exports_gdp	CIV	Côte d'Ivoire	5.079209	4.147575	0.8165790	-4.0559981	4.055998	Exports of goods and services (% of GDP)
exports_gdp	CMR	Cameroon	4.228425	9.232743	2.1834943	-8.4956705	8.495670	Exports of goods and services (% of GDP)
exports_gdp	COG	Republic of Congo	10.891082	12.119244	1.1127677	12.0211839	12.021184	Exports of goods and services (% of GDP)
exports_gdp	GAB	Gabon	9.949549	5.526300	0.5554322	-0.1738516	4.987402	Exports of goods and services (% of GDP)
exports_gdp	MLI	Mali	1.644789	2.337057	1.4208854	0.5028868	1.973068	Exports of goods and services (% of GDP)
exports_gdp	NER	Niger	4.106919	10.403599	2.5331885	-9.9852965	9.985297	Exports of goods and services (% of GDP)
exports_gdp	SEN	Senegal	2.031861	6.559557	3.2283497	-5.6486291	5.648629	Exports of goods and services (% of GDP)
exports_gdp	TCD	Chad	2.778024	20.301452	7.3078740	14.0960344	16.925594	Exports of goods and services (% of GDP)

9 Extension Figures

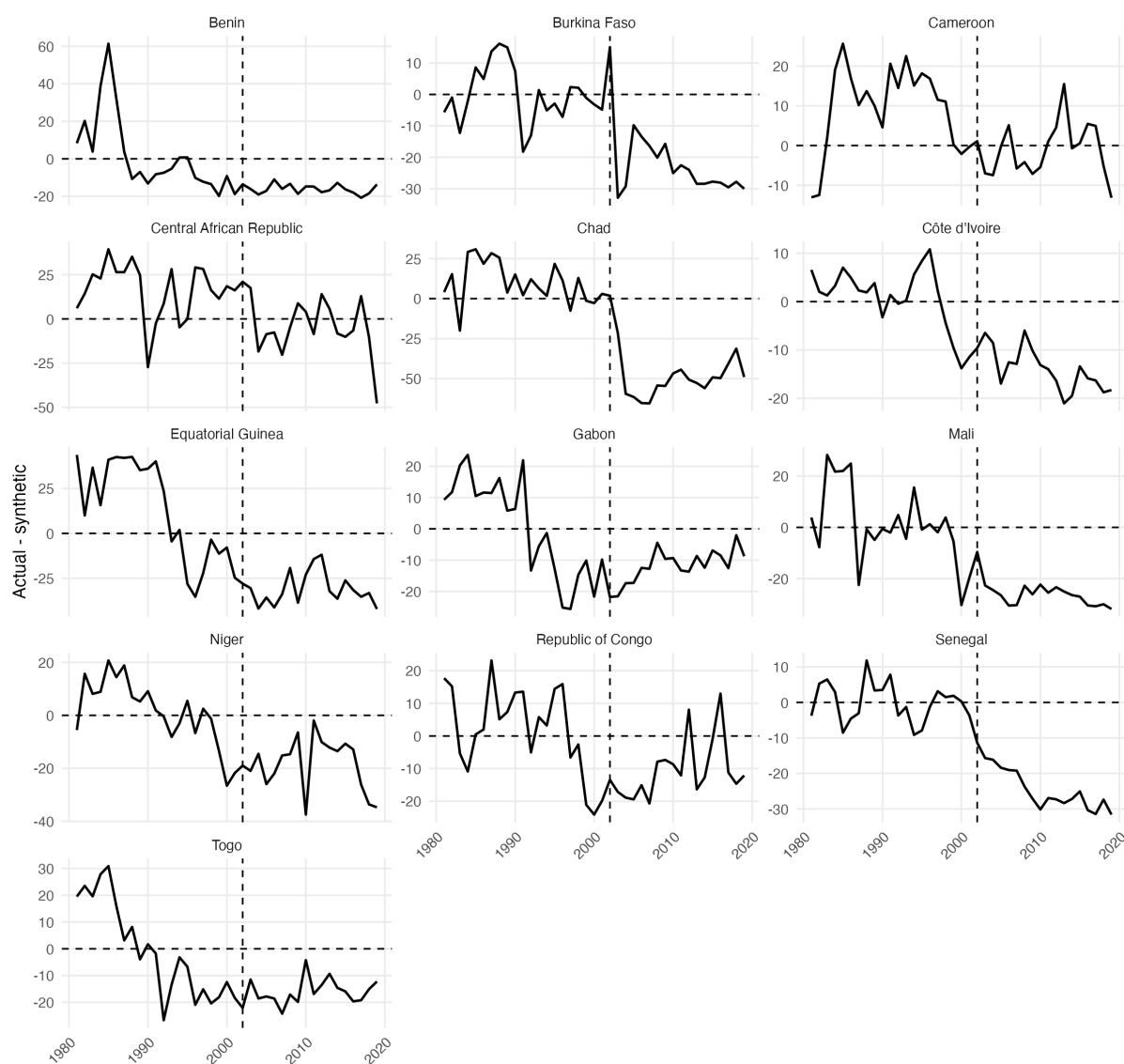
Extension SCM: actual minus synthetic — Imports from EU (% of total imports)

Dashed line = 2002

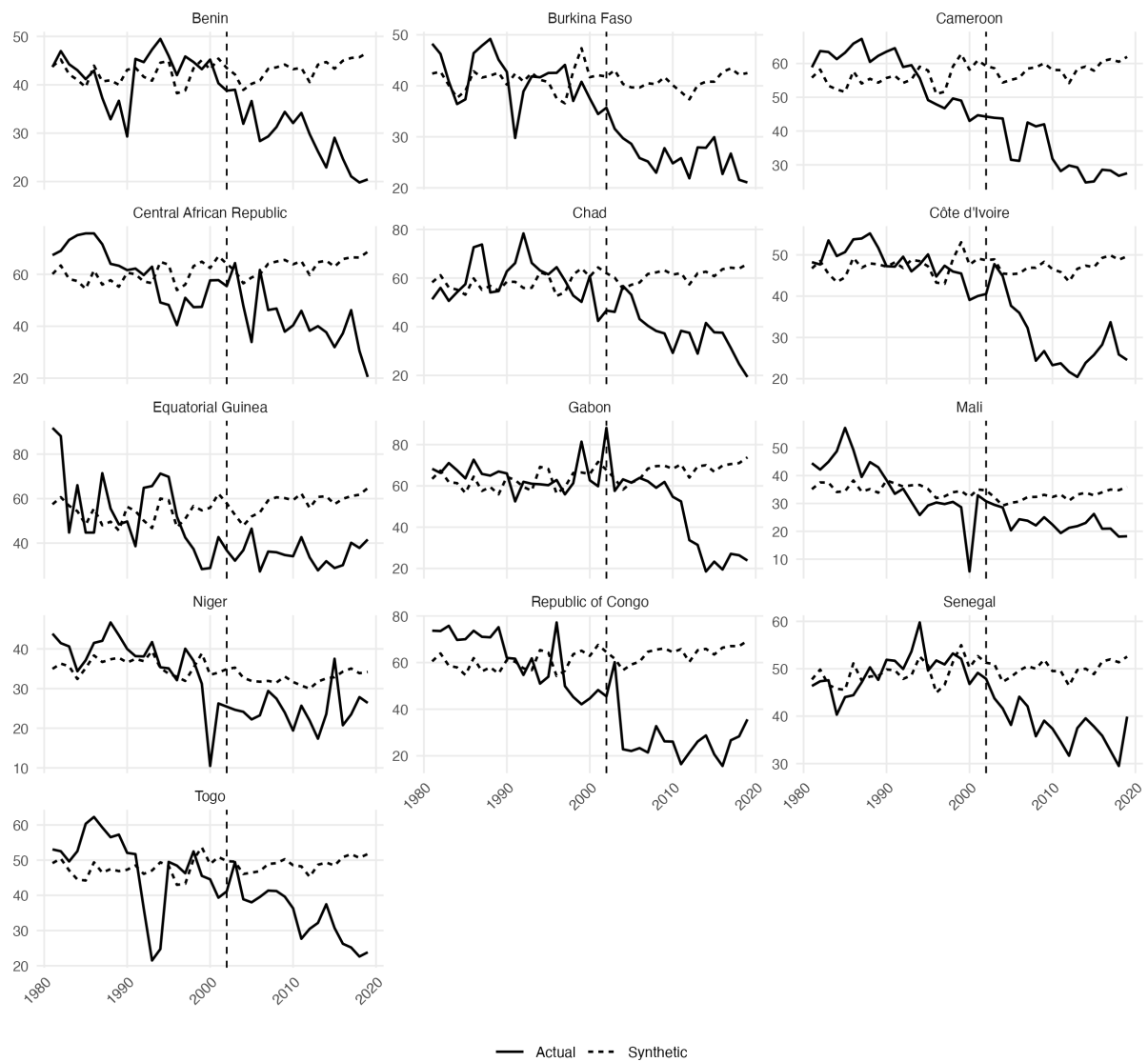


Extension SCM: actual minus synthetic — Exports to EU (% of total exports)

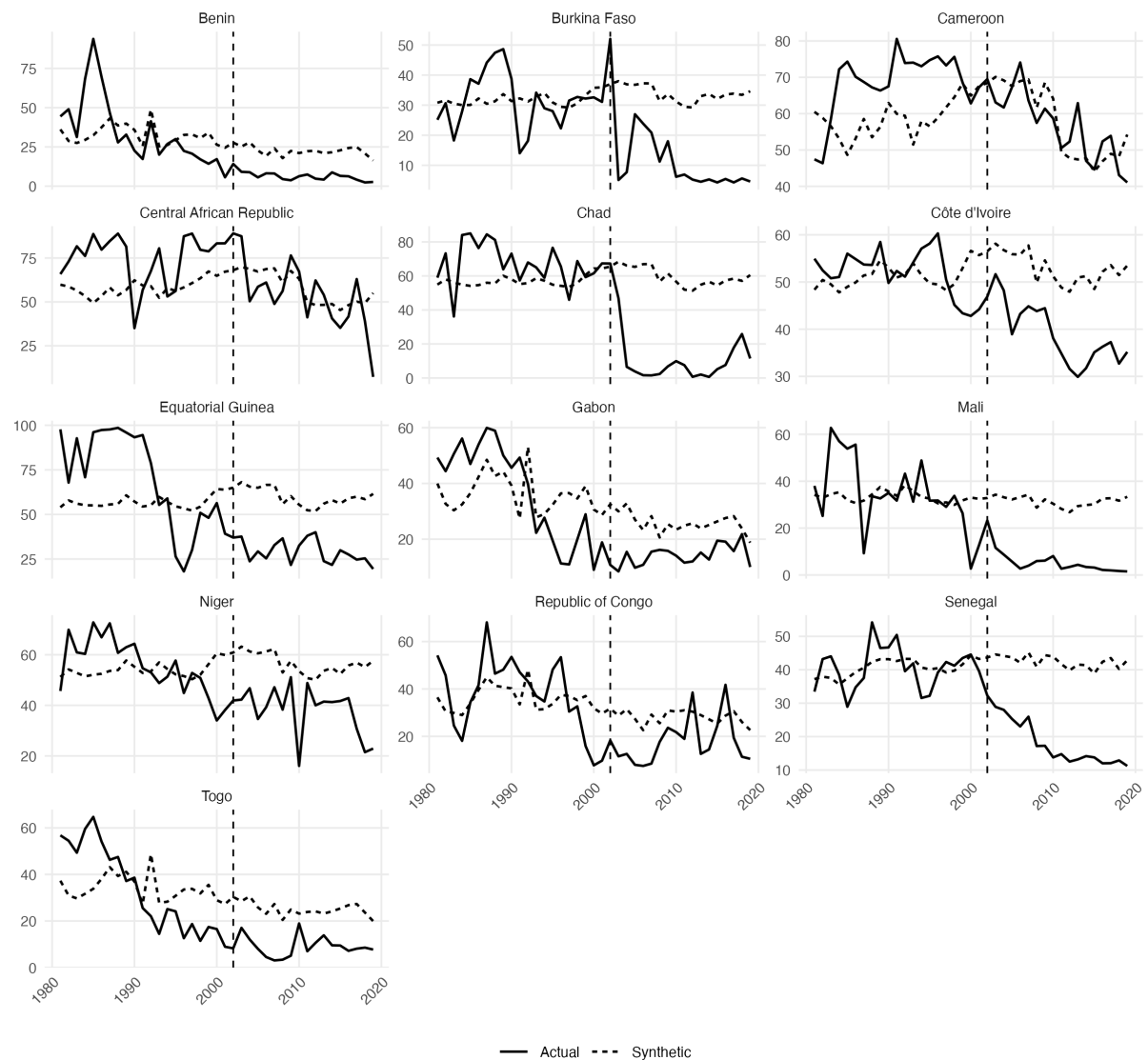
Dashed line = 2002



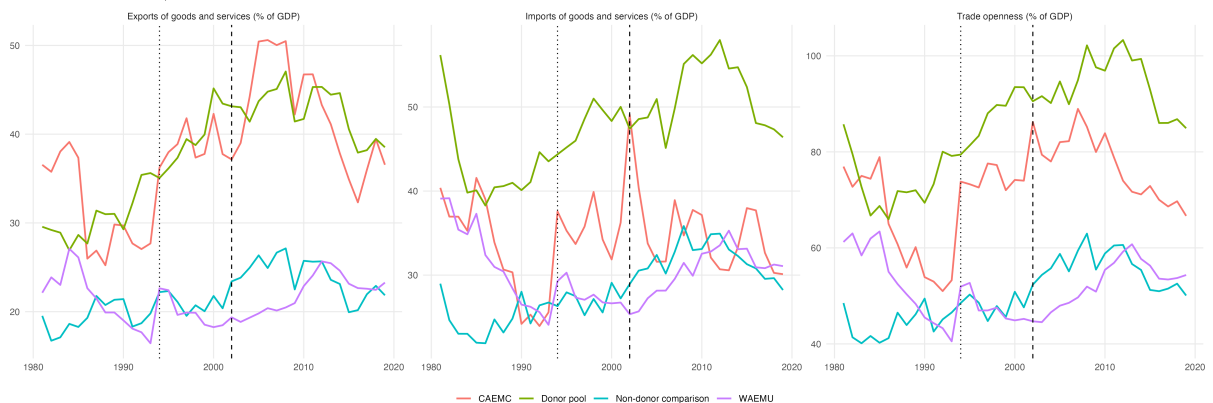
Extension SCM: actual and synthetic — Imports from EU (% of total imports)
Dashed line = 2002



Extension SCM: actual and synthetic — Exports to EU (% of total exports)
Dashed line = 2002



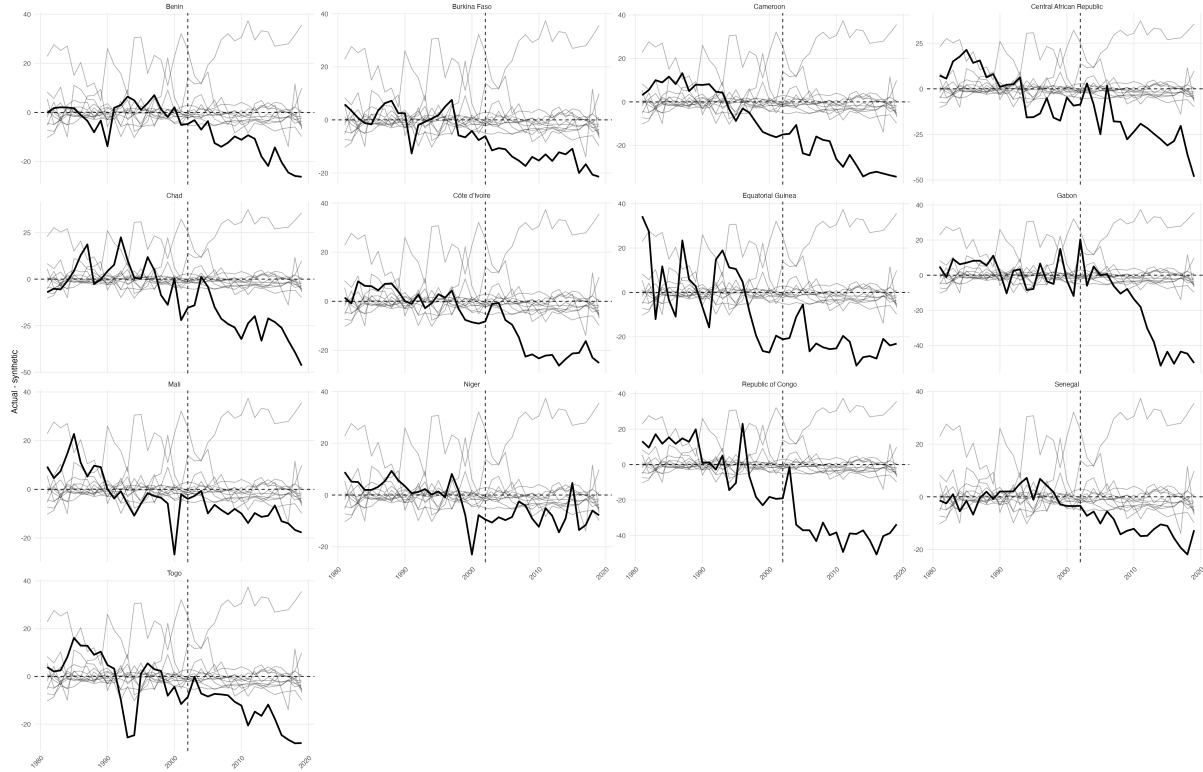
Trade-extension variables: group averages
Dashed vertical line = 2002, Dotted vertical line = 1994



10 Extension Placebo

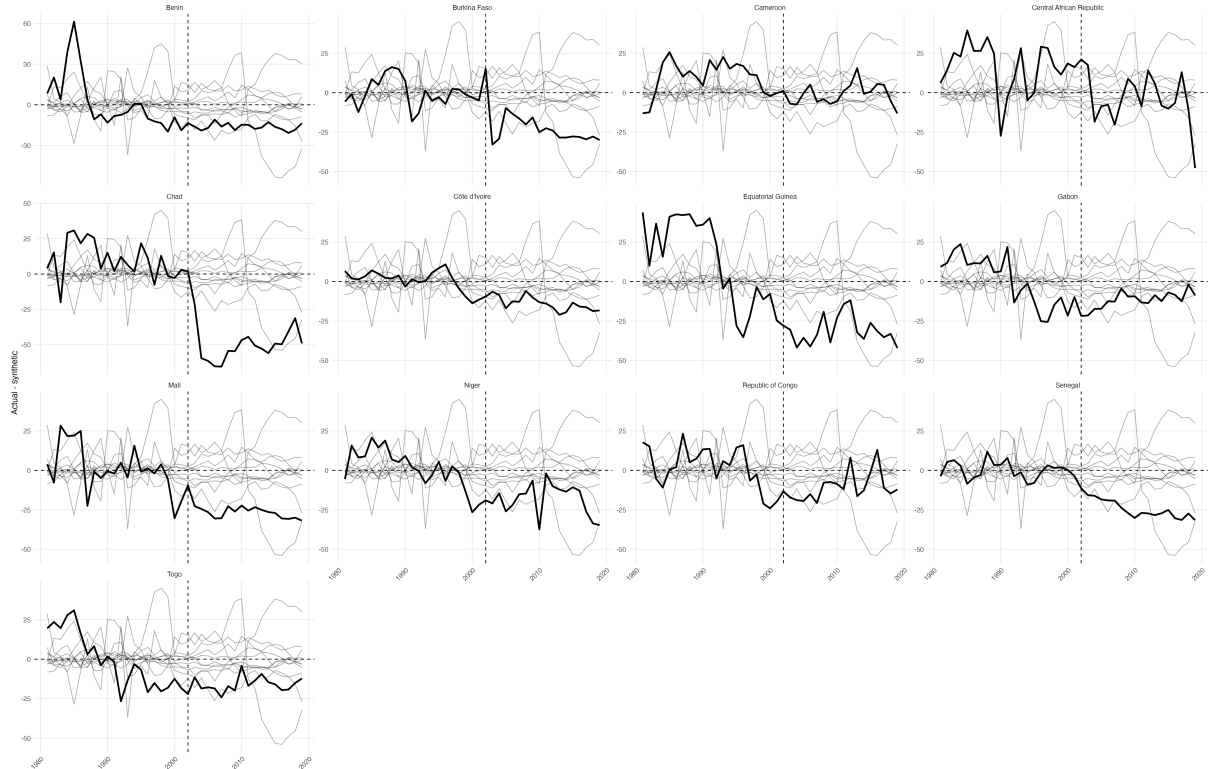
Extension placebo gaps — Imports from EU (% of total imports)

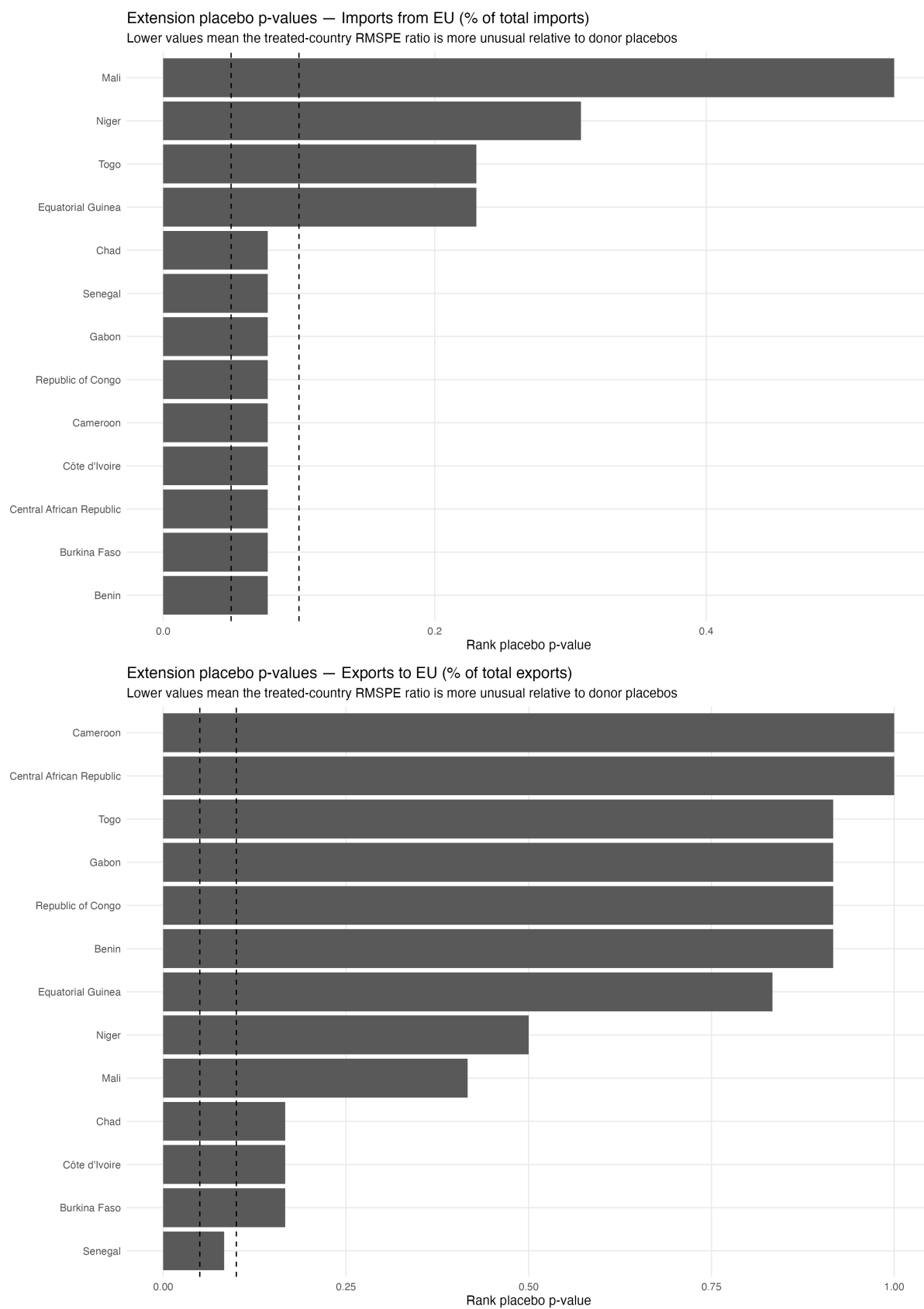
Thin lines = donor placebos; thick line = actual treated country. Dashed vertical line = 2002



Extension placebo gaps — Exports to EU (% of total exports)

Thin lines = donor placebos; thick line = actual treated country. Dashed vertical line = 2002





11 Extension Placebo